

Backend Development of Document Expiry Feature

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Implemented backend support for document expiry management by extending the document data model and adding automated expiry calculations.

BACKEND COMPONENTS

1. Added optional `expiryDate` field to the Document database schema
2. Updated document upload API to accept expiry information
3. Implemented server-side validation for expiry dates
4. Stored expiry metadata securely in the database
5. Developed logic to calculate remaining validity days

TECHNICAL HIGHLIGHTS



DB Schema

Modifications via Prisma ORM



API Layer

Endpoints updated for create & update ops



Date Logic

Server-side calculations for consistency

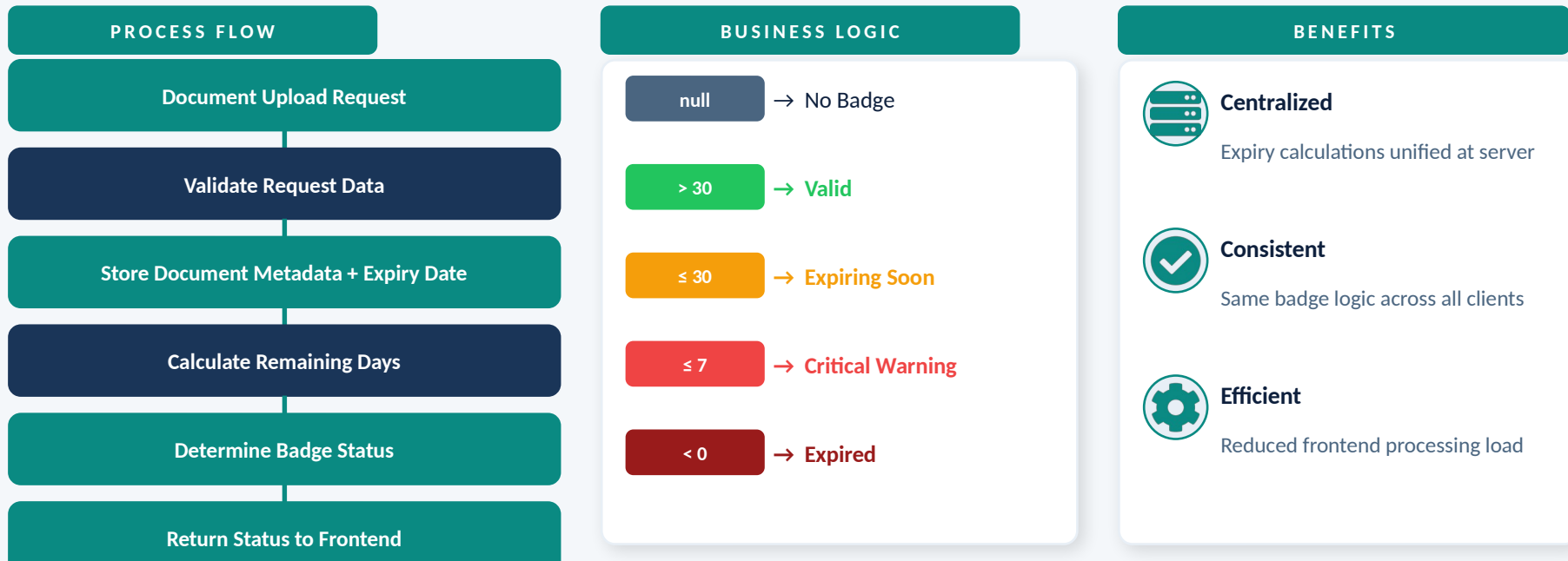


Auto Status

Badge generation from remaining days

Backend Workflow for Expiry Tracking

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All expiry calculations are performed server-side to ensure uniform badge behavior and data integrity across the entire platform.

Backend Development of Report Generation Module

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Developed a server-side report generation system that compiles project information and exports it as a downloadable PDF for Principal Investigators.

BACKEND COMPONENTS



Created Generate Report API endpoint



Aggregated data from multiple project-related tables



Structured project info into a unified report format



Integrated PDF generation library (server-side)



Implemented secure report download functionality

DATA SOURCES

Project Details

Team Information

Documents

Financial Records

Progress Info

TECHNICAL HIGHLIGHTS



Prisma ORM

Dynamic data fetching across tables



Server PDF

Content rendered entirely server-side



Auto Format

Report formatting & org automatic

Backend Workflow for PDF Report Generation

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PROCESS FLOW

Generate Report Request

Authenticate User (PI)

Fetch Project Data

Aggregate Related Records

Build Report Structure

Generate PDF Document

BACKEND ADVANTAGES



Single API

Complete report generation in one endpoint call



Real-Time Data

Live retrieval ensures up-to-date accuracy



Reduced Effort

Manual reporting workload eliminated



Scalable Arch.

Ready for future report enhancements

OUTCOME



Outcome

Principal Investigators can generate comprehensive project reports with a single action. The backend automatically compiles, formats, and exports project data into a professional PDF document.

Scalable, single-API design enables PIs to generate full project reports instantly — from request to download — with no manual data assembly required.